Université de Bordeaux - Master 1 Informatique

Networks and Protocols (Réseaux et Protocoles)

TD 4 – The DNS protocol[1]

The Domain Name System (DNS) translates hostnames to IP addresses, fulfilling a critical role in the Internet infrastructure. In this lab, we'll take a closer look at the client side of DNS. Recall that the client's role in the DNS is relatively simple – a client sends a *query* to its local DNS server, and receives a *response* back. Much can go on "under the covers," invisible to a DNS client, as the hierarchical DNS servers communicate with each other to either recursively or iteratively resolve the client's DNS query. From the DNS client's standpoint, however, the protocol is quite simple – a query is formulated to the local DNS server and a response is received from that server.

From the description of iterative and recursive DNS query resolution, you might think that the local DNS server must be contacted *every* time an application needs to translate from a hostname to an IP address. That's not always true in practice!

Most hosts (e.g., your personal computer) keep a *cache* of recently retrieved DNS records (sometimes called a DNS *resolver cache*), just like many Web browsers keep a cache of objects recently retrieved by HTTP. When DNS services need to be invoked by a host, that host will first check if the DNS record needed is resident in this host's DNS cache; if the record is found, the host will not even bother to contact the local DNS server and will instead use this cached DNS record. A DNS record in a resolver cache will eventually timeout and be removed from the resolver cache, just as records cached in a local DNS server will timeout.

It is possible to explicitly clear the records in one's DSN cache. On a Linux computer, that would be achieved with sudo systemd-resolve –flush-caches.

[1]This activity was adapted from the lab session proposed by Kurose & Ross, v8.1.

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1. Tracing DNS with Wireshark

For this first exercise, you can use the dns-wireshark-trace1-1 trace file. It was obtained by following these steps (which you may replicate if you are working from home):

- Clear the DNS cache in your host.
- Open your Web browser and clear your browser cache.
- Open Wireshark and enter ip.addr == <your_IP_address> into the display filter, where <your_IP_address> is the IPv4 address of your computer[2]. With this filter, Wireshark will only display packets that either originate from, or are destined to, your host.
- Start packet capture in Wireshark.
- With your browser, visit the Web page: http://gaia.cs.umass.edu/kurose_ross/
- Stop packet capture.

Answer the following questions:

1. Locate the first DNS query message resolving the name gaia.cs.umass.edu. What is the packet number[3] in the trace for the DNS query message? Is this query message sent over UDP or TCP?
2. Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?
3. What is the destination port for the DNS query message? What is the source port of the DNS response message?
4. To what IP address is the DNS query message sent?
5. Examine the DNS query message. How many “questions” does this DNS message contain? How many “answers” answers does it contain?
6. Examine the DNS response message to the initial query message. How many “questions” does this DNS message contain? How many “answers” answers does it contain?

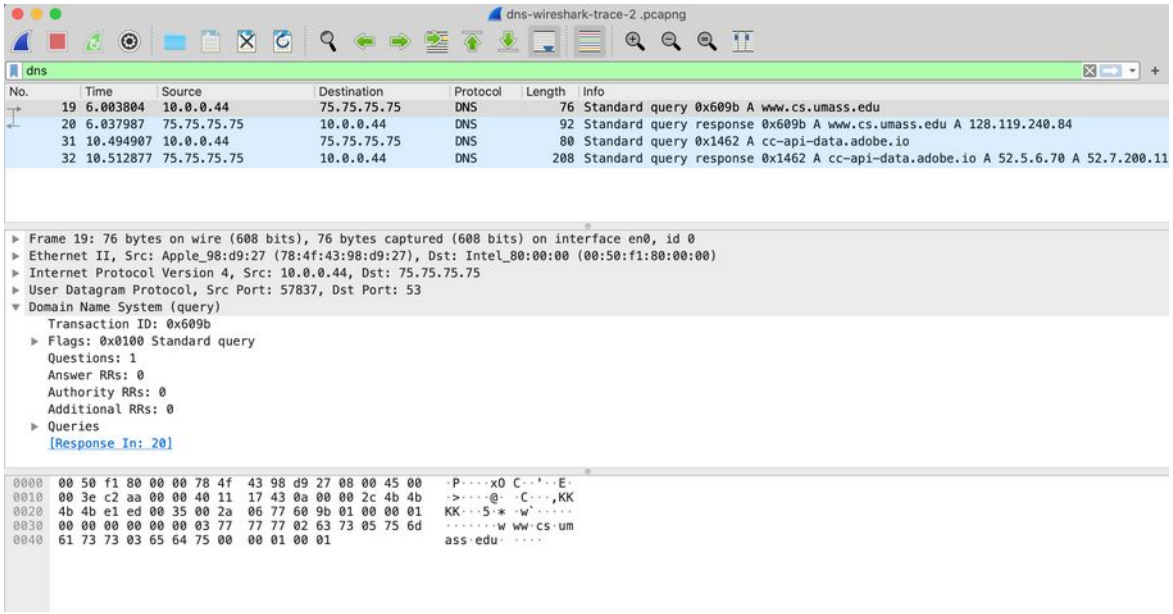
[2] If you’re not sure how to find the IP address of your computer, you can search the Web for articles for your operating system. Windows 10 info is here; Mac info is here; Linux info is here

[3] Remember that this “packet number” is assigned by Wireshark for listing purposes only; it is NOT a packet number contained in any real packet header.

7. The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg, which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address? What is the packet number in the trace of the received DNS response? What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg? What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address? Discuss how DNS caching affects the answer to this last question.

For the next questions, you may use the dns-wireshark-trace-2 file, which was obtained by capturing packets while using the nslookup command, which may be used to query DNS servers. The performed query was to obtain DNS entries related to www.cs.umass.edu.

You should get a trace that looks something like the following in your Wireshark window. Let's look at the first type A query (which is packet number 19 in the figure below, and indicated by the "A" in the *Info* column for that packet.



8. What is the destination port for the DNS query message? What is the source port of the DNS response message?
9. To what IP address is the DNS query message sent? Is this the IP address of the default local DNS server of the person who generated this trace?
10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?
11. Examine the DNS response message to the query message. How many “questions” does this DNS response message contain? How many “answers”?

Last, let’s use nslookup to issue a command that will return a type NS DNS record, that can be done with the following commmand: nslookup –type=NS umass.edu. **The *dns-wireshark-trace-3 trace file contains a trace captured while doing so. Use it to answer the following questions:***

12. To what IP address is the DNS query message sent? Is this the IP address of the default local DNS server of the person obtaining the trace file?
13. Examine the DNS query message. How many questions does the query have? Does the query message contain any “answers”?
14. Examine the DNS response message. How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records?